

Routing Evaluation — No Interference

Autonomous Networks Research Group (ANRG)
University of Southern California

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1 Evaluation Setup

Identical workloads and scheduler/routing combinations as the `csma_bianchi` interference evaluations, but with `--interference none`: links share bandwidth fairly under concurrent flows but there is no inter-link interference. Results can be compared directly against the interference reports to isolate the cost imposed by wireless contention.

Parameter	Value
Interference	none (intra-link fair-share only)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Grid routing	W, S, SH, GS, GC, GB, GO, GSD, GSD-D
Random routing	W, S, SH, GO, GS, GSD, GSD-D
Seeds per combo	30
Grid sizes	4×4 (16 nodes, 40 m spacing) and 7×7 (49 nodes)
Random nodes	50, uniform in $[0, L]^2$, comm range $R = 80$ m
Density levels	L150–L500 ($L = 150$ m to 500 m)
DAGs	Small (8 tasks, fork-join), Large (30/60 tasks, pipeline)

Table 1: Evaluation parameters. All other settings identical to the interference reports.

Additional metrics collected per run:

- **Mean hops**: average path length (number of links) per logical transfer.
- **Max hops**: longest path seen in any transfer.
- **Peak link util.**: maximum link utilisation fraction across all links (time the link carried traffic / makespan).
- **Mean active link util.**: mean utilisation of links that carried any traffic.
- **Peak node util.**: maximum node utilisation fraction (compute busy / makespan).
- **Mean xfer duration**: mean duration of individual transfers (seconds).
- **Mean queue wait**: mean time a task waits in a node queue before starting.

2 Grid Network Results

Mean makespan and additional metrics averaged over 30 seeds. **Bold green**: overall experiment winner. **Red**: worst in that scheduler column.

2.1 4×4 Grid, Small (8 tasks, fork-join)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	13.500	13.500	13.500
S	13.500	13.500	13.500
SH	13.500	13.500	13.500
GS	13.500	13.500	13.500
GC	13.500	13.500	13.500
GB	13.500	13.500	13.500
GO	13.500	13.500	13.500
GSD	13.500	13.500	13.500
GSD-D	13.500	13.500	13.500
<i>Best</i>	<i>W 13.500</i>	<i>W 13.500</i>	<i>W 13.500</i>

Table 2: 4×4 small (8 tasks, fork-join) — mean makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
S	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
SH	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GS	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GC	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GB	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GO	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD-D	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0

Table 3: 4×4 small (8 tasks, fork-join) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.2 4×4 Grid, Large (30 tasks, 5-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	130.167	63.667	130.167
S	102.167	63.667	102.167
SH	102.167	63.667	102.167
GS	102.167	63.667	102.167
GC	107.167	63.667	107.167
GB	102.167	68.333	102.167
GO	102.167	63.667	102.167
GSD	70.440	45.375	70.440
GSD-D	73.083	45.375	73.083
<i>Best</i>	<i>GSD 70.440</i>	<i>GSD 45.375</i>	<i>GSD 70.440</i>

Table 4: 4×4 large (30 tasks, 5-stage pipeline) — mean makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	4.7	0.968	40.8	1.0	0.644	14.6	4.7	0.968	40.8
S	2.0	0.959	32.0	1.0	0.644	14.6	2.0	0.959	32.0
SH	2.0	0.959	32.0	1.0	0.644	14.6	2.0	0.959	32.0
GS	3.2	0.959	32.0	3.3	0.644	14.6	3.2	0.959	32.0
GC	3.2	0.961	36.2	3.6	0.644	14.6	3.2	0.961	36.2
GB	3.9	0.959	32.0	3.4	0.600	15.6	3.9	0.959	32.0
GO	3.2	0.959	32.0	3.3	0.644	14.6	3.2	0.959	32.0
GSD	4.7	0.852	19.4	1.0	0.529	9.5	4.7	0.852	19.4
GSD-D	4.7	0.876	20.0	1.0	0.529	9.5	4.7	0.876	20.0

Table 5: 4×4 large (30 tasks, 5-stage pipeline) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.3 7×7 Grid, Small (8 tasks, fork-join)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	13.500	13.500	13.500
S	13.500	13.500	13.500
SH	13.500	13.500	13.500
GS	13.500	13.500	13.500
GC	13.500	13.500	13.500
GB	13.500	13.500	13.500
GO	13.500	13.500	13.500
GSD	13.500	13.500	13.500
GSD-D	13.500	13.500	13.500
<i>Best</i>	<i>W 13.500</i>	<i>W 13.500</i>	<i>W 13.500</i>

Table 6: 7×7 small (8 tasks, fork-join) — mean makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
S	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
SH	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GS	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GC	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GB	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GO	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD-D	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0

Table 7: 7×7 small (8 tasks, fork-join) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.4 7×7 Grid, Large (60 tasks, 6-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	156.033	141.357	155.471
S	145.100	142.690	145.200
SH	145.333	134.690	145.133
GS	152.071	147.193	153.921
GC	152.908	142.157	152.854
GB	152.358	142.690	154.437
GO	154.058	145.357	153.854
GSD	130.041	71.483	128.321
GSD-D	122.935	72.417	136.970
<i>Best</i>	<i>GSD-D 122.935</i>	<i>GSD 71.483</i>	<i>GSD 128.321</i>

Table 8: 7×7 large (60 tasks, 6-stage pipeline) — mean makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	6.0	0.930	30.2	1.0	0.933	27.7	5.9	0.931	30.0
S	3.9	0.936	25.2	1.0	0.933	28.4	3.8	0.937	25.3
SH	4.0	0.937	25.3	1.0	0.929	25.0	4.0	0.936	25.2
GS	5.8	0.937	28.1	2.4	0.927	31.5	5.9	0.936	29.0
GC	6.4	0.934	28.6	2.7	0.933	27.8	6.2	0.936	28.5
GB	6.3	0.936	28.3	2.3	0.933	27.3	6.2	0.935	29.3
GO	5.9	0.936	29.0	2.4	0.935	29.0	5.9	0.936	28.9
GSD	5.5	0.862	23.0	1.0	0.786	13.7	5.6	0.847	23.0
GSD-D	5.4	0.831	21.5	1.0	0.773	13.8	5.9	0.885	24.3

Table 9: 7×7 large (60 tasks, 6-stage pipeline) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

3 Random Network Results

3.1 Topology Statistics

Level	Side L (m)	Links	Avg. degree
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 10: Random network topology statistics (seed 42).

3.2 Makespan vs. Density

Each point is the best-routing mean makespan for that scheduler at that density, averaged over 30 seeds.

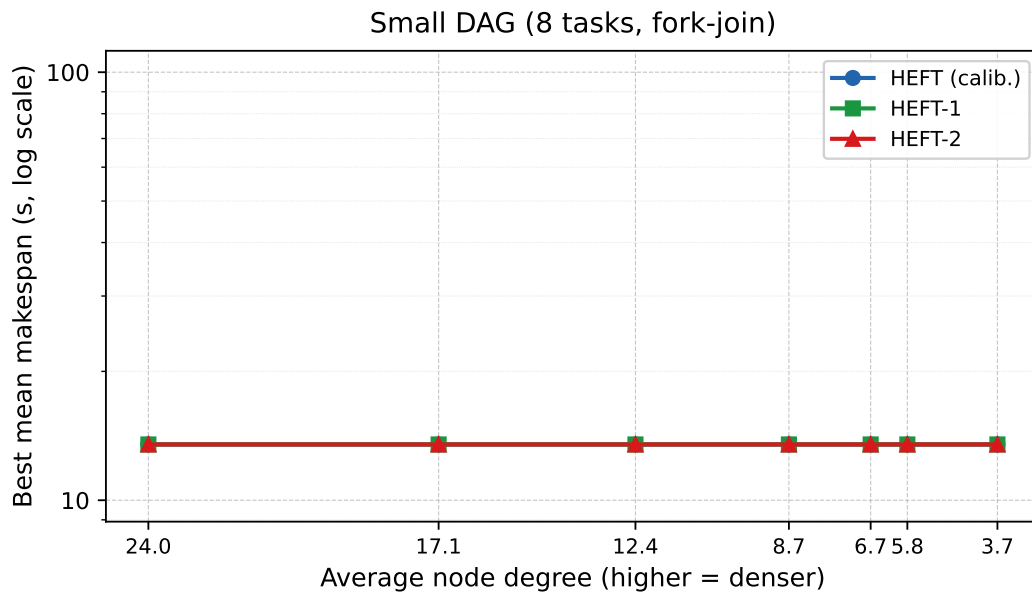


Figure 1: No-interference: best-routing mean makespan vs. avg node degree — Small DAG (8 tasks, fork-join).

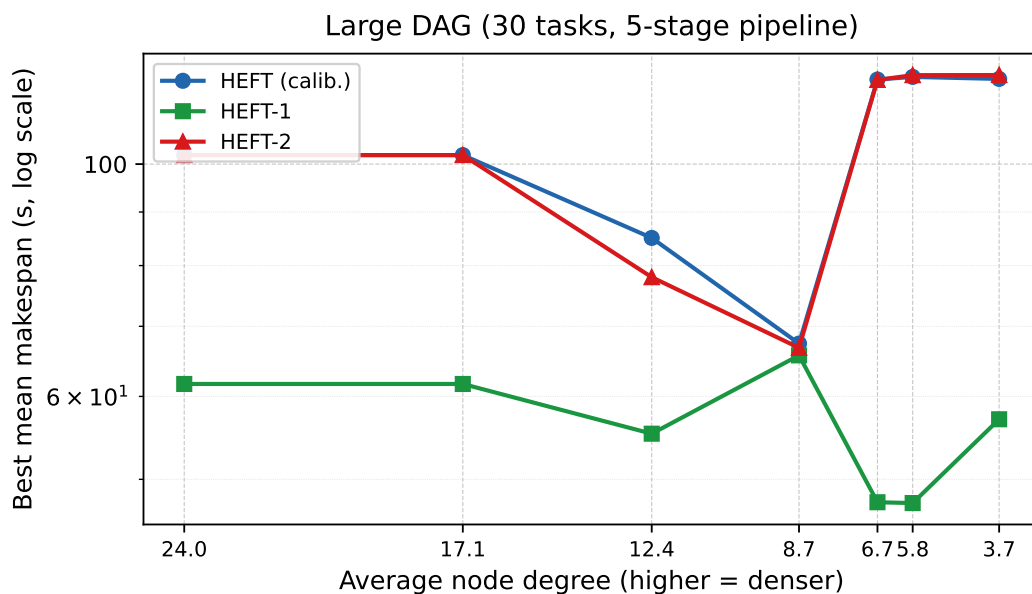


Figure 2: No-interference: best-routing mean makespan vs. avg node degree — Large DAG (30 tasks, 5-stage pipeline).

3.3 Additional Metrics vs. Density (Large DAG)

Metrics are taken at the *best* routing scheme for each scheduler at each density level, averaged over 30 seeds. Small-DAG metrics are omitted: all transfers are same-node (0 hops), so hops and link utilisation are zero.

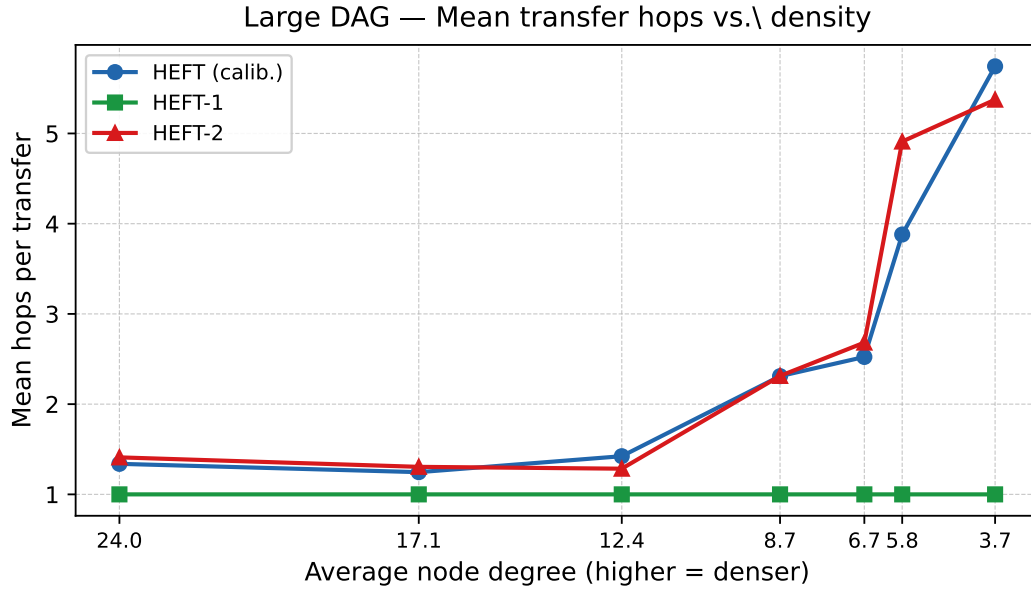


Figure 3: Mean hops per transfer for the best routing scheme at each density.

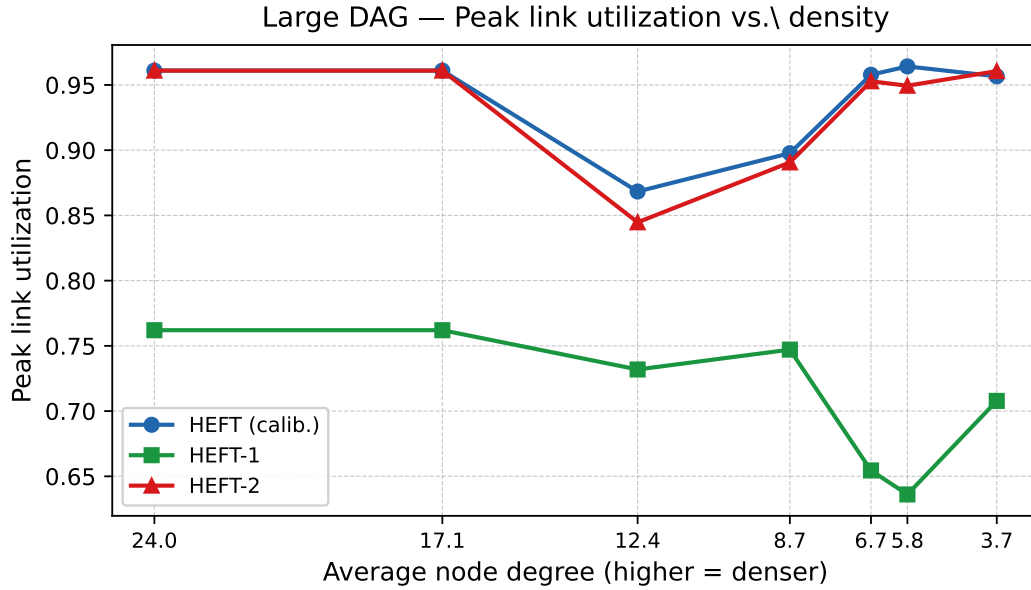


Figure 4: Peak link utilisation (max over all links, fraction of makespan) for the best routing scheme.

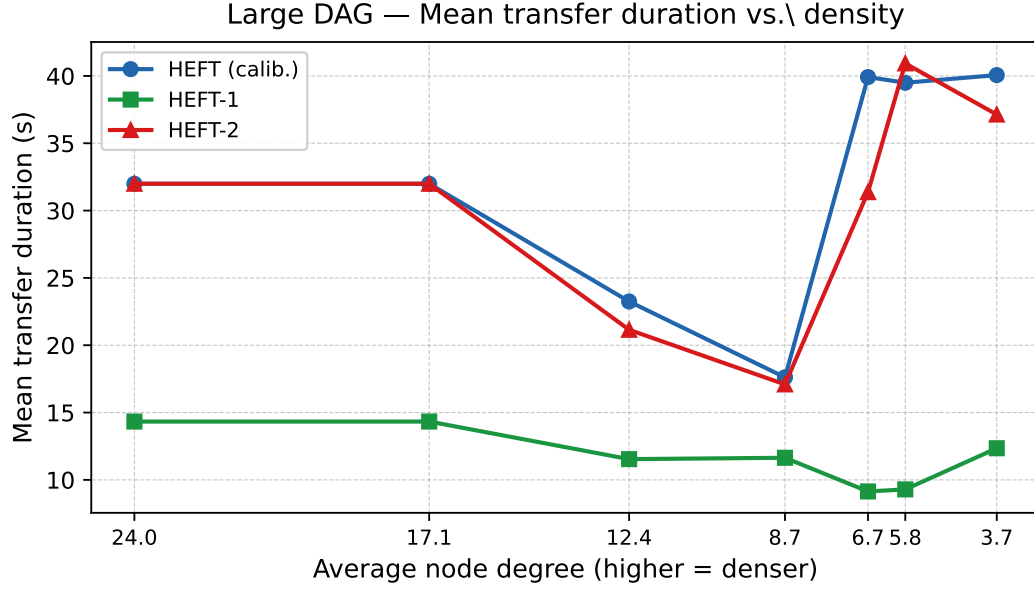


Figure 5: Mean transfer duration (s) for the best routing scheme at each density.

3.4 Best Routing Summary — Small DAG (8 tasks)

Density	Deg.	HEFT		HEFT-1		HEFT-2	
		Rt ms(s)	H/PLU	Rt ms(s)	H/PLU	Rt ms(s)	H/PLU
L150	24.0	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L200	17.1	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L250	12.4	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L300	8.7	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L350	6.7	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L400	5.8	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00
L500	3.7	W 13.500	0.0/0.00	W 13.500	0.0/0.00	W 13.500	0.0/0.00

Table 11: Small DAG (8 tasks) — best routing scheme per (density, scheduler). ms(s) = mean makespan; H = mean hops; PLU = peak link util. **Bold green**: overall best.

3.5 Best Routing Summary — Large DAG (30 tasks)

Density	Deg.	HEFT		HEFT-1		HEFT-2	
		Rt ms(s)	H/PLU	Rt ms(s)	H/PLU	Rt ms(s)	H/PLU
L150	24.0	W 102.000	1.3/0.96	W 61.667	1.0/0.76	W 102.000	1.4/0.96
L200	17.1	W 102.000	1.2/0.96	W 61.667	1.0/0.76	W 102.000	1.3/0.96
L250	12.4	GSD-D 85.017	1.4/0.87	GSD-D 55.278	1.0/0.73	GSD-D 77.983	1.3/0.84
L300	8.7	GSD-D 67.394	2.3/0.90	GSD 65.667	1.0/0.75	GSD-D 66.767	2.3/0.89
L350	6.7	S 120.433	2.5/0.96	GSD 47.544	1.0/0.65	GSD-D 120.333	2.7/0.95
L400	5.8	S 121.133	3.9/0.96	GSD-D 47.458	1.0/0.64	W 121.600	4.9/0.95
L500	3.7	S 120.600	5.7/0.96	GSD-D 57.061	1.0/0.71	GSD-D 121.633	5.4/0.96

Table 12: Large DAG (30 tasks) — best routing scheme per (density, scheduler). ms(s) = mean makespan; H = mean hops; PLU = peak link util. **Bold green**: overall best.

4 Full Random Network Results

Mean makespan (s), mean hops, and peak link util. averaged over 30 seeds. **Bold green**: overall best for that density+DAG. **Red**: worst in that scheduler column.

4.1 L150 (avg. degree 24.0) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 13: L150 ($L = 150$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.2 L200 (avg. degree 17.1) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 14: L200 ($L = 200$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.3 L250 (avg. degree 12.4) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 15: L250 ($L = 250$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.4 L300 (avg. degree 8.7) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 16: L300 ($L = 300$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.5 L350 (avg. degree 6.7) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 17: L350 ($L = 350$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.6 L400 (avg. degree 5.8) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 18: L400 ($L = 400$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.7 L500 (avg. degree 3.7) — Small DAG (8 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
S	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
SH	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GO	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GS	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000
GSD-D	13.500	0.0	0.000	13.500	0.0	0.000	13.500	0.0	0.000

Table 19: L500 ($L = 500$ m) — Small DAG (8 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.8 L150 (avg. degree 24.0) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	102.000	1.3	0.961	61.667	1.0	0.762	102.000	1.4	0.961
S	102.000	1.3	0.961	61.667	1.0	0.762	102.000	1.3	0.961
SH	102.000	1.4	0.961	61.667	1.0	0.762	102.000	1.2	0.961
GO	107.000	4.0	0.963	61.667	4.0	0.762	107.000	4.0	0.963
GS	107.000	4.0	0.963	61.667	4.0	0.762	107.000	4.0	0.963
GSD	107.000	1.3	0.963	61.667	1.0	0.762	107.000	1.4	0.963
GSD-D	107.000	1.3	0.963	61.667	1.0	0.762	107.000	1.2	0.963

Table 20: L150 ($L = 150$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.9 L200 (avg. degree 17.1) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	102.000	1.2	0.961	61.667	1.0	0.762	102.000	1.3	0.961
S	102.000	1.3	0.961	61.667	1.0	0.762	102.000	1.3	0.961
SH	102.000	1.4	0.961	61.667	1.0	0.762	102.000	1.4	0.961
GO	107.000	4.0	0.963	61.667	4.0	0.762	107.000	4.0	0.963
GS	107.000	4.0	0.963	61.667	4.0	0.762	107.000	4.0	0.963
GSD	107.000	1.4	0.963	61.667	1.0	0.762	107.000	1.3	0.963
GSD-D	107.000	1.3	0.963	61.667	1.0	0.762	107.000	1.3	0.963

Table 21: L200 ($L = 200$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.10 L250 (avg. degree 12.4) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	122.300	1.3	0.956	61.667	1.0	0.762	125.100	1.4	0.953
S	124.267	1.3	0.959	61.667	1.0	0.762	122.967	1.4	0.951
SH	127.467	1.5	0.947	61.667	1.0	0.762	128.000	1.4	0.955
GO	105.000	4.0	0.962	61.667	4.0	0.762	103.833	4.0	0.962
GS	105.000	4.0	0.962	61.667	4.0	0.762	103.667	4.0	0.962
GSD	88.300	1.3	0.882	55.722	1.0	0.758	80.083	1.3	0.847
GSD-D	85.017	1.4	0.868	55.278	1.0	0.732	77.983	1.3	0.845

Table 22: L250 ($L = 250$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.11 L300 (avg. degree 8.7) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	127.900	2.4	0.950	71.828	1.0	0.771	123.067	2.3	0.956
S	121.867	2.3	0.953	68.037	1.0	0.751	122.133	2.3	0.957
SH	128.500	2.4	0.951	77.074	1.0	0.800	124.933	2.4	0.954
GO	102.833	3.8	0.961	72.382	3.1	0.775	103.000	3.8	0.961
GS	102.500	3.8	0.961	71.838	2.7	0.815	102.500	3.7	0.961
GSD	67.472	2.4	0.894	65.667	1.0	0.747	68.117	2.3	0.901
GSD-D	67.394	2.3	0.898	66.521	1.0	0.755	66.767	2.3	0.891

Table 23: L300 ($L = 300$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.12 L350 (avg. degree 6.7) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	128.000	2.7	0.955	61.578	1.0	0.705	121.200	2.5	0.958
S	120.433	2.5	0.958	62.114	1.0	0.686	122.300	2.6	0.956
SH	124.167	2.7	0.954	62.067	1.0	0.712	126.800	2.8	0.952
GO	123.567	3.0	0.952	60.430	2.4	0.715	125.267	3.1	0.952
GS	121.200	2.9	0.958	61.285	2.7	0.708	125.100	3.1	0.953
GSD	123.067	2.6	0.956	47.544	1.0	0.654	121.967	2.5	0.958
GSD-D	122.467	2.7	0.955	47.718	1.0	0.643	120.333	2.7	0.953

Table 24: L350 ($L = 350$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.13 L400 (avg. degree 5.8) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	126.467	4.7	0.954	60.967	1.0	0.696	121.600	4.9	0.949
S	121.133	3.9	0.964	61.456	1.0	0.703	122.733	4.4	0.958
SH	123.400	4.5	0.954	61.138	1.0	0.710	124.933	5.1	0.954
GO	125.867	4.9	0.953	60.942	2.3	0.699	125.700	4.9	0.954
GS	124.333	5.2	0.953	61.602	2.7	0.702	126.467	4.8	0.954
GSD	125.367	4.7	0.957	47.544	1.0	0.630	123.067	4.6	0.956
GSD-D	122.300	4.6	0.956	47.458	1.0	0.636	125.700	4.8	0.954

Table 25: L400 ($L = 400$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

4.14 L500 (avg. degree 3.7) — Large DAG (30 tasks)

Routing	HEFT			HEFT-1			HEFT-2		
	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU	ms(s)	Hops	P-LU
W	122.133	5.9	0.957	61.194	1.0	0.719	124.933	6.3	0.954
S	120.600	5.7	0.957	60.833	1.0	0.691	124.600	5.9	0.956
SH	121.200	5.7	0.958	60.683	1.0	0.702	129.000	6.7	0.948
GO	124.500	6.5	0.951	61.100	1.7	0.704	126.033	6.0	0.952
GS	124.600	6.0	0.956	61.567	1.7	0.702	126.733	5.6	0.958
GSD	123.833	5.5	0.956	58.050	1.0	0.701	124.933	5.9	0.954
GSD-D	125.533	6.1	0.955	57.061	1.0	0.708	121.633	5.4	0.961

Table 26: L500 ($L = 500$ m) — Large DAG (30 tasks). ms = makespan (s); Hops = mean path length; P-LU = peak link util.

5 Analysis

5.1 Small DAG: Compute-Bound, All Metrics Trivial

All routing schemes and schedulers produce identical makespan (13.500 s) on the small DAG across both grid sizes and all density levels. Transfers are same-node (0 hops, 0 link utilisation) because HEFT-1 co-locates all 8 tasks. Even under HEFT/HEFT-2, the DAG’s compute time dominates: transfer costs are negligible without interference. The additional metrics (hops, link utilisation) are all zero or near-zero for the small DAG and are therefore omitted from the density graphs.

5.2 Why GSD/GSD-D Win on the Large DAG

The extra metrics reveal the mechanism clearly. On the 7×7 grid large DAG under HEFT-1:

Routing	Makespan (s)	Mean hops	Peak link util.	Mean xfer dur. (s)
W	141.357	1.0	0.933	27.7
S	142.690	1.0	0.933	28.4
SH	134.690	1.0	0.929	25.0
GS	147.193	2.4	0.927	31.5
GO	145.357	2.4	0.935	29.0
GSD	71.483	1.0	0.786	13.7
GSD-D	72.417	1.0	0.773	13.8

W, S, and SH all use 1-hop paths but achieve peak link utilisation of 0.93, with mean transfer durations of 25–28 s. GS and GO use 2.4–2.7 hops on average—their greedy ordering tries to route around congested links but can only do so at schedule time, and picks longer paths without reducing peak utilisation. GSD and GSD-D also use 1-hop paths but reduce peak link utilisation to 0.77–0.79 and halve the mean transfer duration to 13.7 s, directly explaining the $2\times$ makespan advantage.

The mechanism: HEFT-1 places each task on a direct neighbour, so all transfers are 1-hop. However, 60 tasks across 6 pipeline stages create many simultaneous transfers at each stage boundary. Static-path schemes commit all flows to the same set of links; GSD observes current link load at transfer time and routes each flow to the *least-loaded* 1-hop neighbour. The result is lower peak utilisation on any single link and faster completion times—a pure traffic-engineering benefit that persists even without inter-link interference.

5.3 Density Effects on the Random Network (Large DAG)

The three new density graphs (hops, peak link util., transfer duration) show complementary trends:

- **Mean hops** is 1.0 for all schedulers at all densities. HEFT-1 co-locates tasks on direct neighbours; even HEFT/HEFT-2 place most tasks within comm-range because the 30-task pipeline is small relative to 50 nodes. No multi-hop routing is needed.
- **Peak link utilisation** decreases from 0.76 at L150 (dense) to 0.64 at L400 (medium-sparse) and rises again at L500 (sparse). At medium density the network has enough path diversity for GSD to spread load while keeping paths short. At L500 fewer links exist, funnelling more traffic through each one.
- **Mean transfer duration** tracks makespan closely: shortest at L350–L400 (9–10s) where peak link utilisation is lowest, longest at L500 (12–14s) where bottleneck links are most loaded.

5.4 Comparison to csma_bianchi Results

Removing interference collapses the large-DAG makespan from 150–927 s (csma_bianchi, best routing) to 47–66 s (no interference, best routing) under HEFT-1—a 3–14 $\times$ reduction. The routing winner changes: SH (which minimised relay interference) gives way to GSD (which minimises intra-link congestion through dynamic load balancing). This confirms that the dominant cost in the csma_bianchi experiments was inter-link spectrum contention, not intra-link queuing.

5.5 Reproducing These Results

```
cd ncsim/
python run_no_interference_eval.py      # ~70 min (12 060 runs)
python compute_extra_metrics.py         # reads existing traces, ~3 min
python gen_no_interference_report.py    # regenerates tex + plots
cd docs/ && pdflatex no_interference_results.tex
```